



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

12-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 12-06-2021

ANSWER KEY**PAPER-2****PHYSICS**

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B	A	A	A	A	D	A
SEC-2	Q.	1	2	3	4	5	6	7
	A.	B,C	A,B,C	A,B,C	A,B,C	A,B,C	B,C	A,C
SEC-3	Q.	1	2	3	4			
	A.	B	B	A	A			

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B	D	C	C	D	B	A
SEC-2	Q.	1	2	3	4	5	6	7
	A.	C,D	A,D	B,D	A,B,C,D	B	A,B,C,D	A,C,D
SEC-3	Q.	1	2	3	4			
	A.	C	C	A	B			

MATHEMATICS

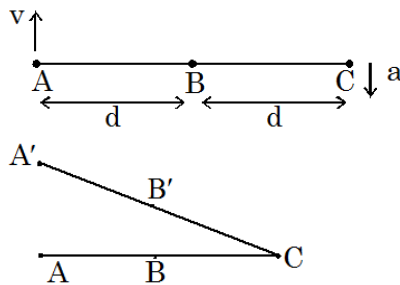
SEC-1	Q.	1	2	3	4	5	6	7
	A.	A	A	D	A	C	A	B
SEC-2	Q.	1	2	3	4	5	6	7
	A.	A,B	B,C,D	A,B	A,B,C	A,B,C	C,D	A
SEC-3	Q.	1	2	3	4			
	A.	C	C	A	B			

HINT – SHEET

PHYSICS

SEC-1

2. Ans (A)



Let us observe the motion of A and B relative to C.

$$\frac{AA'}{2d} = \frac{BB'}{d}$$

$$BB' = \frac{AA'}{2} = \frac{vt + \frac{1}{2}at^2}{2}$$

$$BB' = \frac{Vt}{2} + \frac{1}{4}at^2$$

$\frac{V}{2}$ is initially speed of B w.r.t. C as well as ground.

$\frac{a}{2}$ is acceleration of B w.r.t. C

$$\vec{a}_{B/g} = \vec{a}_{B/C} + \vec{a}_{C/g}$$

$$= \frac{a}{2} - a = -\frac{1}{2}a$$

4. Ans (A)

$$I_1 = \frac{I_A}{2}$$

$$m = (\sigma a^2)$$

$$m_1 = \sigma(2a)^2 = 4m$$

$$m_2 = \sigma(\sqrt{2}a)^2 = 2m$$

$$I_A = \left(\frac{m_1(2a)^2 + m_1a^2}{6} \right) - \left(\frac{m_2(\sqrt{2}a)^2 + m_2a^2}{6} \right)$$

$$I_A = 4ma^2 \left(\frac{2+1}{3} \right) - 2ma^2 \left(\frac{1}{3} + 1 \right)$$

$$= ma^2 \left[\frac{20-8}{3} \right] = 4ma^2$$

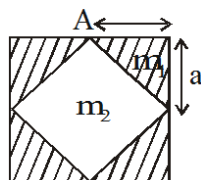
$$I_1 = 2ma^2$$

$$I_2 = \frac{ma^2}{6} + m \left(\frac{a}{\sqrt{2}} \right)^2$$

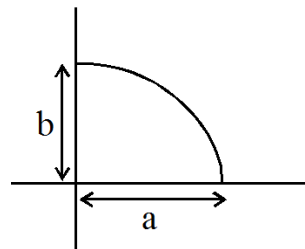
$$I_2 = ma^2 \left(\frac{1}{6} + \frac{1}{2} \right) \Rightarrow \frac{ma^2}{2} \cdot \frac{4}{3}$$

$$I_2 = \frac{2ma^2}{3}$$

$$\frac{I_1}{I_2} = 3$$



5. Ans (A)



$$x = \frac{\int dm x}{\int dm} = \frac{\int dv x}{\int dv}$$

$$\int dv = \int \pi y^2 dx; \frac{y^2}{b^2} = \frac{1-x^2}{a^2}$$

$$= \pi b^2 \int \left(\frac{1-x^2}{a^2} \right) dx$$

$$= \pi b^2 \left[\frac{x-x^3}{3a^2} \right]_0^a$$

$$= \pi b^2 \left(\frac{a-a}{3} \right) = \frac{2\pi}{3} b^2 a$$

$$\int dvx = \int \pi y^2 dx x$$

$$= \pi b^2 \int \left(1 - \frac{x^2}{a^2} \right) x dx$$

$$= \pi b^2 \left(\frac{x^2}{2} - \frac{x^4}{4a^2} \right)_0^a$$

$$= \pi b^2 \left(\frac{a^2}{2} - \frac{a^4}{4} \right) = \mu b^2 \frac{a^2}{4}$$

$$\left(X_{cm} = \frac{\frac{\pi b^2 a^2}{4}}{\frac{2\pi}{3} b^2 a} = \frac{3}{8} a \right)$$

6. Ans (D)

$$mv = -mv_1 + Mv_2$$

$$t = \frac{d}{v} + \frac{d}{ev} + \frac{d}{e^2v} + \dots$$

$$t = \frac{d}{v} \left(1 + \frac{1}{e} + \frac{1}{e^2} + \dots \right)$$

Common ratio > 1 so $t \rightarrow \infty$

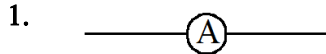
7. Ans (A)

$$\sqrt{r} = (Z-b) \left(1 - \frac{1}{n^2} \right)^{1/2}$$

PHYSICS

SEC-2

Ans (B,C)



$$R_A = 100\Omega$$

$$I_f = 50 \times 10^{-6}$$

$$(a) V_{\max} = 50 \times 10^{-6} [100 + 10 \times 10^3]$$

$$V_{\max} \cong 50 \times 10^{-6} \times 10^4 \cong 0.5V$$

$$(b) V_{\max} = 50 \times 10^{-6} [100 + 200 \times 10^3]$$

$$\cong 50 \times 10^{-6} \times 200 \times 10^3 \cong 10V$$

$$(c) 50 \times 10^{-6} \times 100 = x \times 5 \times 10^{-3}$$

$$x = 10A \quad I_{\max} \cong 10A$$

$$(d) 50 \times 10^{-6} \times 100 = x$$

$$x = 5 \times 10^{-6}A \quad I_{\max} = 55 \times 10^{-6}A$$

2. Ans (A,B,C)

Charges on the inner and outer plates will be

$$\frac{Q_1 + Q_2}{2} \left| \left(\frac{Q_1 - Q_2}{2} \right) - \left(\frac{Q_1 - Q_2}{2} \right) \right| \frac{Q_1 + Q_2}{2}$$

$$q = \left(\frac{\theta_1 - \theta_2}{2} \right) e^{-t/RC}$$

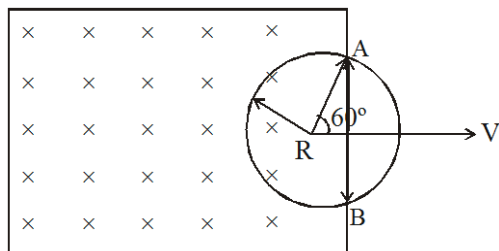
3. Ans (A,B,C)

At any time t

$$I = \frac{\varepsilon}{\lambda 2\pi R} = \frac{B(2AB)v}{\lambda 2\pi R}$$

$$I = \frac{Bv}{\lambda \pi R} \sqrt{R^2 - \left(\frac{R}{2} - vt \right)^2}$$

$$I = \frac{Bv}{\lambda \pi} \sqrt{1 - \left(\frac{1}{2} - \frac{vt}{R} \right)^2}$$



$$I = \frac{2Bv}{\lambda 2\pi R} \sqrt{R^2 - \frac{R^2}{4} + Rvt - v^2 t^2}$$

$$I = \frac{Bv}{\lambda 2\pi R} \sqrt{3R^2 + 4Rvt - 4v^2 t^2}$$

5. Ans (A,B,C)

$$g_{\text{eff}} = \sqrt{g^2 + \left(\frac{qE}{m} \right)^2}$$

$$T \cos \theta = m g_{\text{eff}}$$

$$T \cos \theta = m R \omega^2$$

$$\frac{R \omega^2}{g_{\text{eff}}} = \tan \theta$$

$$\omega^2 = g_{\text{eff}} \frac{\tan \theta}{\ell \sin \theta} \Rightarrow \frac{g_{\text{eff}}}{\ell \cos \theta}$$

$$\omega = \sqrt{\frac{g_{\text{eff}}}{\ell \cos \theta}}$$

$$\cos \theta = \frac{I}{g_{\text{eff}}}$$

$$\sin \theta = \frac{qE}{m g_{\text{eff}}}$$

Time period

$$= \frac{2\pi}{\omega} = \frac{2\pi}{g_{\text{eff}}} \sqrt{\ell g} = 2\pi \sqrt{\frac{\ell g}{g^2 + \left(\frac{qE}{m} \right)^2}}$$

$$v = R\omega = \ell \sin \theta \sqrt{\frac{g_{\text{eff}}}{\ell \cos \theta}}$$

$$v = \ell \frac{qE}{m g_{\text{eff}}} \cdot \sqrt{\frac{g_{\text{eff}}^2}{\ell g}}$$

$$v = \frac{qE}{m} \cdot \sqrt{\frac{\ell}{g}}$$

$$T = \frac{m g_{\text{eff}}^2}{g} \Rightarrow \frac{m}{g} \left[g^2 + \left(\frac{qE}{m} \right)^2 \right]$$

$$T = \left[mg + \frac{q^2 E^2}{mg} \right]$$

6. Ans (B,C)

$$kx - f = Ma$$

$$f = Ma'$$

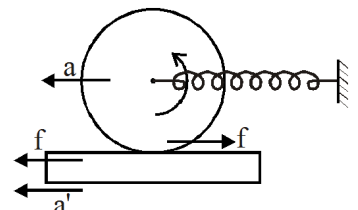
$$a' = a - R\alpha$$

$$fR = \frac{1}{2} MR^2 a$$

$$\frac{f}{M} = a - \frac{2f}{M}$$

$$\frac{3f}{M} = a$$

$$a' = \frac{3f}{M} - \frac{2f}{M} = \frac{f}{M}$$



7. **Ans (A,C)**

Intensity is maximum when the area over which the power is distributed is minimum i.e. at the point 1.

Frequency is max. when relative velocity of approach is maximum.

PHYSICS

SEC-3

1. **Ans (B)**

$$\Delta x = \mu_1 SS_2 + \mu_2 S_2 P$$

$$- \left[\mu_1 SS_1 + \mu_2 (S_1 P - t) + \int \mu_3 dx \right]$$

from graph $\int \mu_3 dx = \text{Area}$

$$= \frac{1}{2}(1+3)t \Rightarrow 2t$$

$$\Delta x = \mu_1 SS_2 + \mu_2 S_2 P$$

$$- [\mu_1 SS_1 + \mu_2 S_1 P - \mu_2 t + 2t]$$

In order to get central maxima at centre

$$\Delta x = 0$$

$$\Delta x =$$

$$- \mu_1 [SS_2 - SS_1] + \mu_2 [S_2 P - S_1 P] + \mu_2 t - 2t$$

$$SS_2 - SS_1 = \sqrt{D^2 + d^2} - D$$

$$\Rightarrow D \left[1 + \frac{d^2}{D^2} \right]^{1/2} - D$$

$$= D \left[1 + \frac{1}{2} \frac{d^2}{D^2} \right] - D$$

$$SS_2 - SS_1 = \frac{1}{2} \frac{d^2}{D}$$

$$\Delta x = \mu_1 \left(\frac{1}{2} \frac{d^2}{D} \right) + \mu(0) + \mu_2 t - 2t$$

$$2t - \mu_2 t = 2 \times \frac{1}{2} \frac{(10^{-3})^2}{(1\text{m})}$$

$$2t - \frac{3t}{2} = 10^{-6}$$

$$\frac{t}{2} = 10^{-6}$$

$$t = 2 \times 10^{-6} \text{ m } (t = 2\mu\text{m})$$

2. **Ans (B)**

$$\Delta x = \mu_1 [SS_2 - SS_1] + \mu_2 [S_2 P - S_1 P] + \mu_2 t$$

$$- 2t$$

For central maxima

$$0 = 2 \frac{1}{2} \frac{d^2}{D} + \frac{3}{2} (S_2 P - S_1 P) + \frac{3t}{2} - 2t$$

$$\frac{3}{2} (S_2 P - S_1 P) = 2t - \frac{3t}{2} - \frac{d^2}{D}$$

$$(S_2 P - S_1 P) = \frac{2}{3} \left[\frac{t}{2} - 10^{-6} \right]$$

$$\frac{2}{3} \left[\frac{10^{-6}}{2} - 10^{-6} \right]$$

$$S_2 P - S_1 P \Rightarrow -\frac{1}{3} \mu\text{m}$$

$$d \sin \theta \cong d \tan \theta$$

$$d \frac{y}{D} = -\frac{1}{3} \mu\text{m}$$

$$y = -\frac{1}{3} \frac{10^{-6} \times 1}{10^{-3}} \text{m}$$

$$y = -\frac{1}{3} \text{mm}$$

3. Ans (A)

$$\eta_{th} = \frac{\text{Net workdone}}{\text{Net heat added}}$$

Since processes 1-2 and 3-4 are adiabatic processes, the heat transfer during the cycle takes place only during processes 2-3 and 4-1 respectively. Therefore, thermal efficiency can be written as,

$$\eta_{th} = \frac{\text{Heat added} - \text{Heat rejected}}{\text{Heat added}}$$

Consider 'm' kg of working fluid,

$$\text{Heat added} = mC_V (T_3 - T_2)$$

$$\text{Heat Rejected} = mC_V (T_4 - T_1)$$

$$\eta_{th} = \frac{mC_V(T_3 - T_2) - mC_V(T_4 - T_1)}{mC_V(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

For the reversible adiabatic processes 3-4 and 1-2, we can write,

$$\frac{T_4}{T_3} = \left(\frac{v_3}{v_4}\right)^{\gamma-1} \text{ and } \frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{\gamma-1}$$

$$v_2 = v_3 \text{ and } v_4 = v_1$$

$$\frac{T_4}{T_3} = \frac{T_1}{T_2} = \frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

$$\eta_{th} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

The ratio $\frac{V_1}{V_2}$ is called as compression ratio, r.

$$\eta_{th} = 1 - \left(\frac{1}{r}\right)^{\gamma-1}$$

$$\text{Net work done} = mC_V \{(T_3 - T_2) - (T_4 - T_1)\}$$

$$\text{Displacement volume} = (V_1 - V_2)$$

$$= V_1 \left(1 - \frac{1}{r}\right) = \frac{mRT_1}{P_1} \left(\frac{r-1}{r}\right)$$

$$= \frac{mC_V(\gamma-1)T_1}{P_1} \left\{\frac{r-1}{r}\right\}$$

$$\text{Since, } R = C_V (\gamma - 1)$$

4. Ans (A)

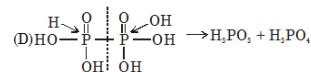
$$\begin{aligned} mep &= \frac{mC_V[(T_3 - T_2) - (T_4 - T_1)]}{\frac{mC_V(\gamma-1)T_1}{P_1} \left\{\left(\frac{r-1}{r}\right)\right\}} \\ &= \left(\frac{1}{\gamma-1}\right) \left(\frac{P_1}{T_1}\right) \left(\frac{r}{r-1}\right) \{(T_3 - T_2) - (T_4 - T_1)\} \end{aligned}$$

CHEMISTRY

SEC-2

3. Ans (B,D)

- (A) $(\text{Mg}(\text{OH})_2 + \text{H}_2\text{C} = \text{C} = \text{CH})$ are produced through a non-redox reaction
(B) N of NH_3 does not have vacant orbital.
(C) $(\text{H}_3\text{BO}_3 + \text{NH}_3 + \text{H}_2\text{O})$ are produced when $\text{B}_2\text{N}_2\text{H}_6$ react with water on heating.



4. Ans (A,B,C,D)

- (A) CO_2 is produced
(B) SO_2 is produced
(C) CO & CO_2 are produced
(D) NO_2 and O_2 are produced.

MATHEMATICS

SEC-1

1. Ans (A)

Let $(\lambda, \lambda, \lambda)$ point of intersection then $\lambda(\sin A + \sin B + \sin C) = 2d^2$ and

$$\lambda(\sin 2A + \sin 2B + \sin 2C) = d^2$$

$$\text{dividing } \frac{\sum \sin 2A}{\sum \sin A} = \frac{1}{2}$$

$$\Rightarrow \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{1}{16}$$

2. Ans (A)

$$(1 + y^2 e^{-2x}) dy = y^3 e^{-2x} dx$$

$$\frac{dy}{y} + y e^{-2x} dy - y^2 e^{-2x} dx = 0$$

Now integrate

3. Ans (D)

$$g(x) = f(x^2 - 2x - 1) + f(5 - x^2 + 2x)$$

$$= 2x^4 - 8x^3 - 4x^2 + 24x + 18$$

$$g'(x) = 8x^3 - 24x^2 - 8x + 24 = 0 \Rightarrow x = -1, 1, 3$$

$$g(x) \geq \min\{g(-1), g(1), g(3)\} = 0$$

$$g(x) \geq 0 \forall x \in \mathbb{R}$$

4. Ans (A)

$\vec{d} = x\vec{a} + y\vec{b} + z\vec{c}$ take dot product with

$$\vec{b} \times \vec{c}, \vec{c} \times \vec{a}, \vec{a} \times \vec{b}$$

$$\vec{d} = \frac{[\vec{d} \vec{b} \vec{c}]\vec{a} + [\vec{d} \vec{c} \vec{a}]\vec{b} + [\vec{d} \vec{a} \vec{b}]\vec{c}}{[\vec{a} \vec{b} \vec{c}]}$$

5. **Ans (C)**

$$(2a)^2 + b^2 = \sin^2 x, (2b)^2 + a^2 = \cos^2 x \text{ where } 3a \text{ and } 3b \text{ legs of triangle then hypotenuse}$$

$$= 3\sqrt{a^2 + b^2} = \frac{3\sqrt{5}}{5}$$

6. **Ans (A)**

$$\text{Equation of directrix is } x - 2y = 0$$

7. **Ans (B)**

$$A^2 = I \text{ then } \begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}^2 = I$$

$$\Rightarrow a = -1 \text{ hence } |A| = -1$$

$$A^{-1} = A \Rightarrow |2A| = -8 \text{ hence } |3\text{adj}.2A|$$

$$= 27(-8)^2 = 1728$$

MATHEMATICS

SEC-2

1. **Ans (A,B)**

$$\text{Let circle } (x-h)^2 + (y-k)^2 = r^2$$

$$r = \left| \frac{ah + bk + 1}{\sqrt{a^2 + b^2}} \right| \Rightarrow a^2(h^2 - r^2) + b^2(k^2 - r^2) +$$

$$2abhk + 2ah + 2bk + 1 = 0$$

$$\text{comparing, we get centre } (3, 0), r = \sqrt{5}$$

2. **Ans (B,C,D)**

$$AM > GM$$

$$2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3 \left(2^{\frac{\pi}{2} + \tan^{-1}x} \right)^{1/3}$$

$$2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3$$

No solution.

3. **Ans (A,B)**

differentiable w.r.t. x,

we get

$$\frac{(mx \cos 4x - \ell \sin 4x)}{x^2} = \frac{\ell(4x \cos x - \sin 4x)}{x^2}$$

$$m = 4\ell$$

Put $m = 4\ell$ to get

$$\ell \int_0^x \left(\frac{4 \cos 4t}{t} - \frac{\sin 4t}{t^2} \right) dt = \frac{\ell \sin 4x}{x} - 1$$

$$4 \int_0^x \frac{\cos 4t}{t} dt + \int_0^x \sin 4t \left(-\frac{1}{t^2} \right) dt = \frac{\sin 4x}{x} - \frac{1}{\ell}$$

$$\text{By parts } \frac{\sin 4x}{x} - 4 = \frac{\sin 4x}{x} - \frac{1}{\ell} \Rightarrow \ell = \frac{1}{4}$$

$$m = 1$$

4. **Ans (A,B,C)**

Any tangent $y = mx \pm 4\sin\theta\sqrt{-m}$ it will pass

through centre of circle if (1, 1) satisfy this

$$m^2 + 2[8\sin^2\theta - 1]m + 1 = 0$$

$$D < 0 \Rightarrow \sin^2\theta < \frac{1}{4}$$

$$\theta \in \left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \frac{7\pi}{6}\right) \cup \left(\frac{11\pi}{6}, 2\pi\right)$$

6. **Ans (C,D)**

$$A = \left| 2 \int_0^1 \sqrt{1-x} dx \right| + \left| 2 \int_{-1}^0 \sqrt{x+1} dx \right| - (\sqrt{2})^2 = \frac{2}{3}$$

$$B = 2 \left(\frac{1}{2} \times 1 - \int_0^1 x^2 dx \right) = \frac{1}{3}$$

7. **Ans (A)**

$$x^2 - \frac{2p}{p-5}x + \frac{p-4}{p-5} = 0$$

$$f(0) > 0, f(2) < 0, f(3) > 0$$

MATHEMATICS

SEC-3

1. **Ans (C)**

No. of subsets having any of first $k - 1$ elements $= 2^{k-1}$

2. **Ans (C)**

No. = total no. – subsets having even integers

3. **Ans (A)**

Total points generated = 20, two selected $= {}^{20}C_2$

No. of points with all odd co-ordinates = 4

No. of points with x odd, y even = 6

No. of points with x even, y odd = 4

No. of points with both even = 6

required number of points

$$= {}^4C_2 + {}^6C_2 + {}^4C_2 + {}^6C_2 = 42$$

$$P = \frac{42}{190}$$

4. **Ans (B)**

If S is the reduced sample space then $n(S) = {}^5C_2 \times 4 = 40$; as there are four vertical lines. On each line we need to choose points such that their mid-point belong to set S only. So, we choose both y-coordinates odd or even.

Since, $y \in \{0, 1, 2, 3, 4\}$

$$\text{So, } n(E) = 4 \times (1 + {}^3C_2) = 16$$

$$\Rightarrow P(E) = \frac{16}{40} = \frac{2}{5}$$